

Sustainable Rocket Redesign

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Executive Summary

This project explores how sustainable/efficiency-driven engineering solutions can assist in the reduction of aerodynamic loads on rocket ships. For this specific project, SpaceX's Starship was used for testing as it was made with human-centered design in mind, a concept prevalent in humanitarian engineering. The primary objective of this project was to reduce the vehicle's ballistic coefficient, increasing structural and thermal margins during reentry while maintaining reasonable mass and aerodynamic performance assumptions.

A simplified model of the top of Starship was modeled in SolidWorks and subsequently modified to change the design in accordance with sustainability-focused engineering principles (eg. mass reduction, aerodynamic efficiency). These modifications were then incorporated into a reduced-order descent simulation using Python and the AeroSandbox Library. The simulation modeled velocity, altitude, and dynamic pressure of the redesigned Starship against the original. The simulation estimates a 35% decrease in peak dynamic pressure during the rocket's descent, the measurements driven by a reduction in mass and an increase in the drag coefficient.

Overall, the results demonstrate how humanitarian engineering practices can theoretically be used to optimize designs and significantly reduce reentry loads without complex modeling.

The Redesign

a. Logic

Two major concerns when analyzing a rocket's successful descent are the ship's dynamic pressure ($q = \frac{1}{2}\rho V^2$) and convective heating ($Q \propto \rho^{1/2}V^3$). Referencing these equations, the atmospheric density and vehicle velocity have to be lowered to enable a safe descent. The two values are directly correlated to the ballistic coefficient, being a smaller value β will cause the vehicle to decelerate at higher altitudes and decrease dynamic pressure and peak convective heating.

Looking at the equation for the ballistic coefficient ($\beta = \frac{m}{C_d A}$), the mass, drag coefficient, and aerodynamic reference area can be changed to improve the ballistic coefficient. For this project, only the mass and drag coefficient were modified, as changing the aerodynamic reference area seemed an inappropriate modification for this model.

These modifications are in line with major humanitarian engineering concepts. Humanitarian engineering emphasizes reducing risk to human life, environmental sustainability, and designing to reduce resource use and long-term costs. For a rocket, especially SpaceX's Starship whose main objective is to enable interplanetary transport for humans, humanitarian engineering is more important than ever.

b. Decreasing Mass

Mass reduction is not pursued to maximize performance, but rather to improve resource efficiency and risk maximization. This redesign focuses on eliminating unnecessary mass while maintaining safety margins for a vehicle of these needs.

1. Structural Optimization

- a. Humanitarian Rationale: Unoptimized structures create wasted material and energy.
- b. Redesign Approach
 - i. Remove excess material from areas that have low thermal flux or low bending loads during descent
 - ii. Apply sufficient safety factors while keeping fail-safe margins.
- c. Impact: Reduces material use, lowers manufacturing emissions due to less needed parts
- d. Mass Reduction: ~3,000 - 4,000 kg

2. Humanitarian Material Choices

- a. Humanitarian Rationale: Reduce environmental impact without impacting safety.

- b. Redesign Approach
 - i. Replace thick stainless steel sections with thinner steel or aluminum-lithium alloys in low-heat regions where it would be acceptable.
 - ii. Prioritize recyclable materials and materials that require less energy to manufacture
 - c. Impact: Greener materials used, lowers manufacturing emissions
 - d. Mass Reduction: ~2,000 - 3,000 kg
- 3. Thermal Protection System Optimization
 - a. Humanitarian Rationale: TPS mass should be proportional to risk but still protective.
 - b. Redesign Approach
 - i. Use maximum protection where necessary and use lighter TPS elsewhere
 - ii. Remove subsequent attachment hardware from removed TPS
 - c. Impact: Maintaining failure risk, reduced internal mass
 - d. Mass Reduction: ~1,500 - 2,000 kg
- 4. Simplification of Non-Essential Systems
 - a. Humanitarian Rationale: Simpler systems fail less often.
 - b. Redesign Approach
 - i. Remove redundant systems not required for safe return
 - ii. Remove overly complex structures
 - c. Impact: Decrease weight while keeping present systems
 - d. Mass Reduction: ~1,000 - 1,500 kg

Combining these optimizations, we get a total decrease in mass of 10,000 kg. From research, the top half of Starship has a dry mass of 120,000 kg. For future calculations, values of 120,000 kg for the baseline Starship and 110,000 kg for the sustainable Starship will be used.

c. Increasing Drag Coefficient

Increasing the drag coefficient is a humanitarian design choice focused on reducing the severity of rocket reentry into the atmosphere. A higher drag coefficient allows for a slower descent at higher altitudes, lessening both thermal and structural loads.

- 1. Larger Aerodynamic Flaps
 - a. Humanitarian Rationale: Create passive drag and slows down the ship.
 - b. Redesign Approach
 - i. Increase surface area of flaps
 - ii. Favor drag-producing shapes

- c. Impact: Allows for the vehicle to slow earlier.
 - d. Drag Coefficient Increase: +0.05 - +0.08
- 2. Blunter Nose Geometry
 - a. Humanitarian Rationale: Lowers aerodynamic heating to protect passengers while saving material.
 - b. Redesign Approach
 - i. Increase nose radius to distribute loads over a larger surface area.
 - ii. Reduce sharp geometry to minimize concentrated heating.
 - c. Impact: Decreases likelihood of TPS cracking during reentry and allows for vehicles to slow earlier.
 - d. Drag Coefficient Increase: +0.05 - +0.08
- 3. Flared Bottom of Ship
 - a. Humanitarian Rationale: Efficient use of resources to increase drag coefficient.
 - b. Redesign Approach
 - i. Create a flared bottom (a curved expansion at the rear of the ship).
 - c. Impact: Airflow separates from the vehicle earlier and creates more drag.
 - d. Drag Coefficient Increase: +0.03 - +0.05

Combining these optimizations, we get a total increase of +0.15 in the drag coefficient. To estimate the baseline drag coefficient of Starship other spacecrafts were referenced such as the Apollo capsule with an estimated drag coefficient of ~1.2 -1.4. Since the Apollo capsule has a blunter geometry than SpaceX's Starship, a safe assumption of $C_d = 1.1$ was used. For future calculations, values of 1.1 for the baseline Starship and 1.25 for the sustainable Starship will be used.

The Simulation

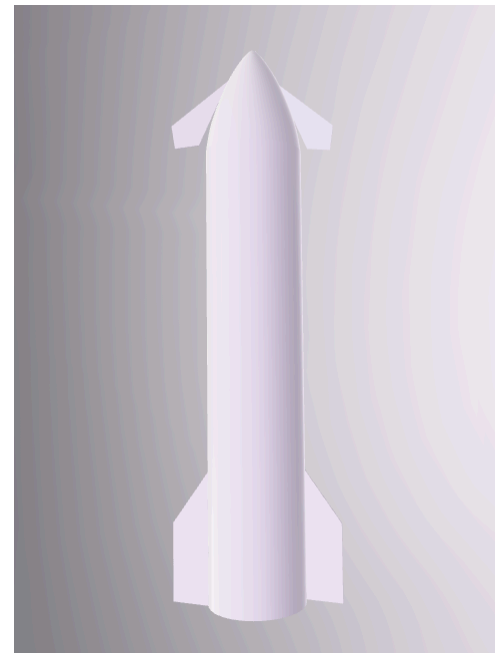
a. Setup

To start, several libraries were imported to facilitate the simulation.

- **AeroSandbox**: The main library used for the simulation.
- **AeroSandbox NumPy**: A version of NumPy to be used alongside AeroSandbox.
- **Trimesh**: Imports the geometry of the STL file to obtain needed values.
- **Matplotlib**: Used to visualize the results.

The first step taken was to create a simplified model of Starship. This model (pictured on the right), was made in SOLIDWORKS. A mesh of the STL file was then loaded into Python using the *trimesh.load* function.

Although the STL file was made to closely model SpaceX's Starship, AeroSandbox was used as a secondary measure to ensure the dimension and measurement of Starship were accurate. To start, values for surface area, volume, and the three dimensions were taken using built in AeroSandbox commands, including *mesh.area*, *mesh.volume*, and the bounding box extents. Using the bounding box, the length of the Starship and the aerodynamic reference area were obtained using the z-direction of the bounding box and the equation $A = \pi(\frac{D}{2})^2$.



Surface area: 2038.2 m²

Volume: 4108.7 m³

Reference area: 1048.9 m²

Length scale: 52.00 m

From the simulation, these values are given. According to official measurements from SpaceX and approximations from other sources, these measurements appear to be close to the correct dimensions.

b. Parameters

To start the parameters, several values had to be established.

- g (gravity) was set to $9.81 \frac{m}{s^2}$ to represent gravity on/near Earth
- dt (change in time) was set to $0.1 s$ so the simulation did not crash from integrating too much
- h_o (initial altitude) was set to $30,000 m$ to represent an altitude in the atmosphere but not high enough to be in orbit
- v_o (initial velocity) was set to $0 \frac{m}{s}$ to represent the start of free-fall

Together, these parameters establish initial conditions to allow the simulation to effectively model the descent of the rocket under gravity's influence.

To compare the baseline and sustainable ships, parameters for masses and drag coefficient were established using the previously estimated values.

c. Physics

To simulate the descent of the ship, a function with the parameters mass, drag coefficient, and reference area was used. To start, arrays of time, altitude, velocity, and dynamic pressure were initialized to store the time history of these values. Using a for loop, these values are repeatedly collected until either the rocket reaches the ground ($h = 0 m$) or the previously set maximum value of time is reached to prevent program crashes. To create an approximation of the rocket's descent, acceleration is used to create an explicit Euler integration scheme.

$$a = g - \frac{D}{mass} \quad \text{with} \quad D = \frac{1}{2} \rho V^2 C_d A$$

Using these equations of Newton's 2nd Law and the drag equation. Out of the values present, we have all of them except for density. Density was calculated using a built-in feature of the AeroSandbox library.

$$\begin{aligned} atm &= asb.Atmosphere(altitude=h) \\ rho &= atm.density() \end{aligned}$$

The first line runs the *asb.Atmosphere* command, an atmospheric model built into AeroSandbox that provides realistic values for an arbitrary object at a certain altitude. The second line extracts the value of density from the atmospheric model to be used in the drag equation. Updating this value each time step is critical since drag scales linearly with density and exponentially with altitude. Now that acceleration is

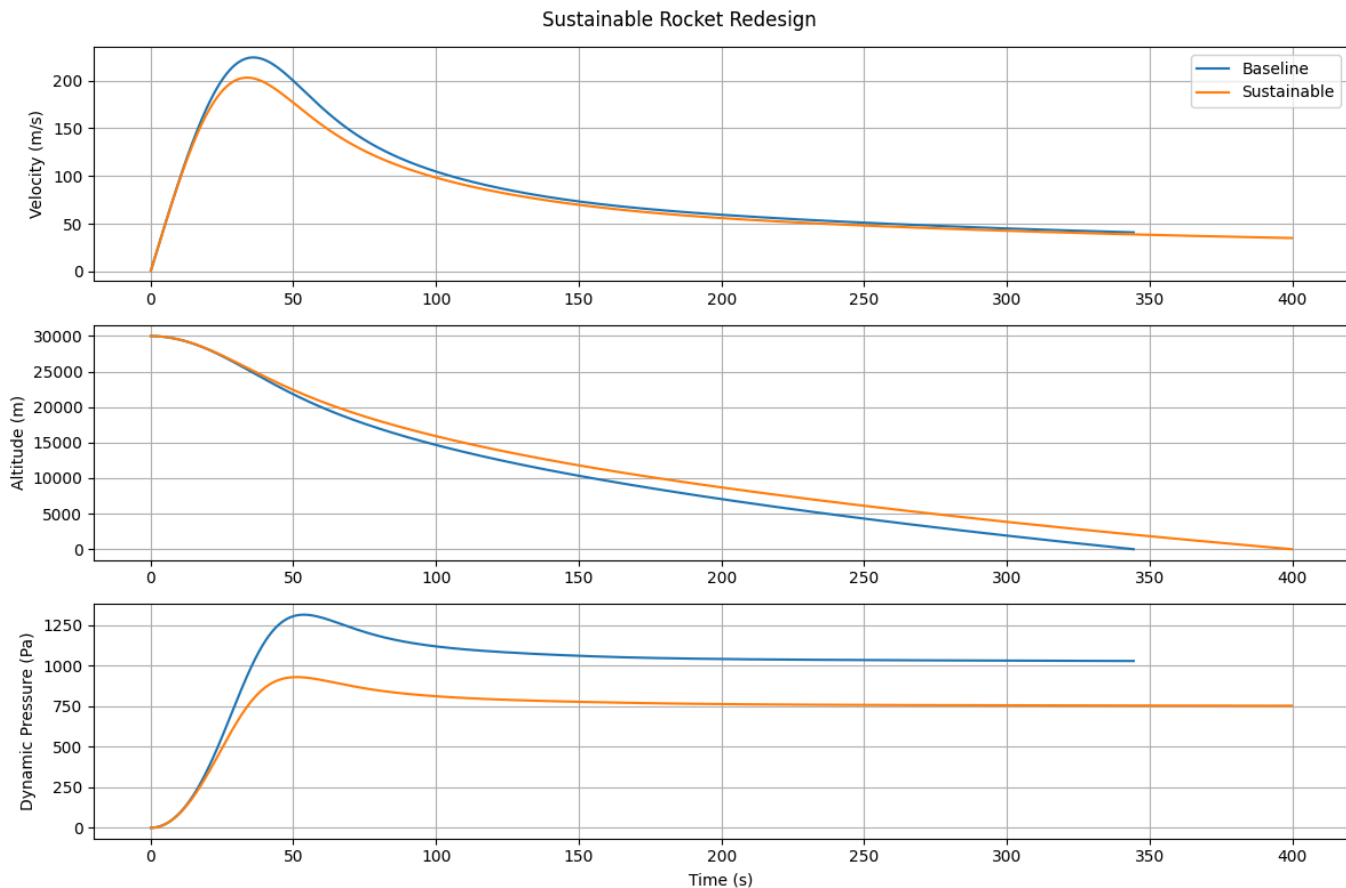
calculated, values for time, altitude, velocity, and dynamic pressure can be calculated.

```
v += a * dt          # Update velocity by adding acceleration over timestep
h -= v * dt          # Update altitude based on velocity over the timestep
t += dt              # Advance simulation by one timestep (0.1 s)
q = 0.5 * rho * v**2 # Compute dynamic pressure from density and velocity
```

The first three lines use Euler integration to approximate the values for each timestep. The last line calculates dynamic pressure using the equation $q = \frac{1}{2}\rho V^2$. Appending these values to their corresponding arrays finishes an iteration of the loop of the function.

With the function programmed, the simulation can now be run. The baseline simulation uses the approximated values of baseline mass, drag coefficient, and reference area of the original Starship. The “sustainable” simulation uses the approximated values of sustainable mass, drag coefficient, and a modified reference area to account for geometric changes of Starship. A 10% increase to the reference area was made to estimate the previously made changes, specifically from the larger aerodynamic flaps and flared bottom.

Results



The velocity plot shows how both vehicles initially accelerate as they begin their descent through the atmosphere. The baseline Starship reaches a maximum velocity of 225 m/s, while the sustainable Starship peaks around 200 m/s. During deceleration, the sustainable redesign maintains a lower velocity throughout its descent, demonstrating the effectiveness of increasing drag coefficient and reference area. This helps to slow the vehicle earlier and more gradually, reduces the ship's aerodynamic and thermal loads.

The altitude plot also highlights the sustainable vehicle's slower descent compared to the baseline. Starting from the same altitude (30,000 m), the baseline Starship reaches the ground after 340 seconds while the sustainable Starship remains airborne for 400 seconds. This indicates the sustainable vehicle brakes earlier in its descent and spreads loads from deceleration over a longer period of time, reducing peak stresses on the vehicle.

The difference in dynamic pressure is apparent through the graph, the baseline experiencing around 1300 Pa at peak dynamic pressure while the sustainable peaks at 950 Pa. This reduction of 27% is coupled by a broader and flatter dynamic pressure curve in the sustainable ship, showcasing how aerodynamic loads are distributed over a longer timeframe.

Conclusion and Limitations

This project investigated how humanitarian engineering principles could be utilized to reduce aerodynamic and structural loads during a rocket's descent. By focusing on reducing ballistic coefficient through reducing mass and increasing drag coefficient, a sustainable redesign caused a lower peak velocity, extended descent time, and substantial decrease in dynamic pressure. These results suggest that these humanitarian design changes had meaningful effects on reentry loads. From a humanitarian engineering perspective, this redesign increases safety margin and reduces risk. Slower deceleration will distribute loads more evenly and lower peak stresses on the vehicle's structure and internal systems. These changes align with the humanitarian goal of designing systems that are safer for transporting life.

However, several limitations must be acknowledged. Firstly, this simulation is a highly simplified representation of an actual rocket descent. This model assumes a vertical trajectory and does not account for lift, angle of attack, or the internal controls of the rocket. In reality, SpaceX's Starship reenters at a high angle of attack and controls its descent using its body flaps, significantly altering the forces applied and the values obtained from this simulation.

Second, this simulation treated the drag coefficient as a constant value. Although this is a good approximation for a simplified model, several factors contribute to a fluctuating drag coefficient value, including Mach number, surface roughness, and most notably the Reynolds Number. The simulation also does not model real thermal behavior and instead uses dynamic pressure to measure aerodynamic loads on the rocket. All structural, material, and TPS mass reductions made in the simulation were done conceptually and were not validated through stress analysis or thermal modeling.

Lastly, it is important to acknowledge that professional engineers already incorporate these considerations at a far more rigorous level than this report details. Real design involves high-fidelity computational dynamics, thermal analysis, finite element analysis, and extensive live testing. The methods explored in this project are very fundamental aspects of the engineering design process.

In conclusion, this project demonstrates that reducing ballistic coefficient through lower mass and higher drag can improve rocket descent safety by decreasing velocity and aerodynamic loads. Although the model is simplified, it effectively shows how basic design choices impact performance and supports the importance of prioritizing safety in engineering.